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Trading futures and ETFs

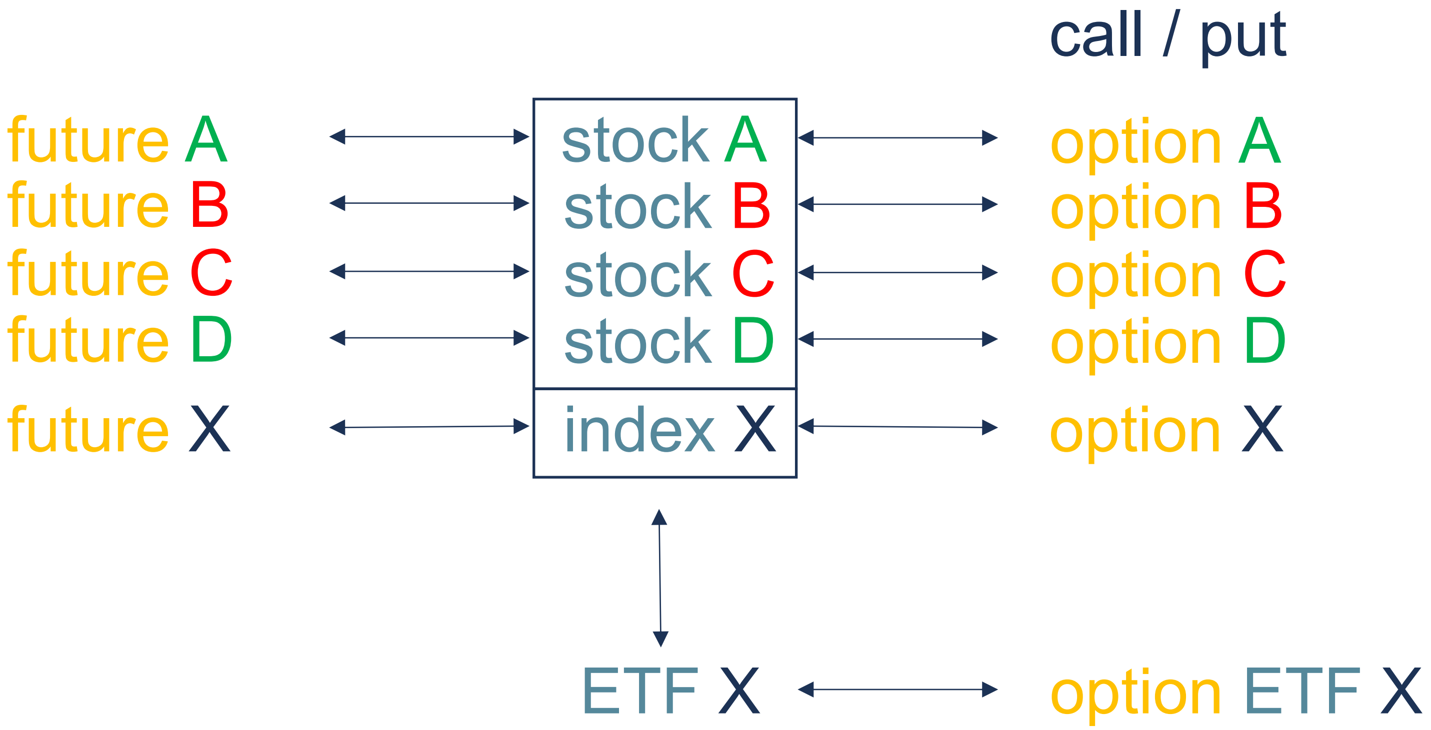
Robbert Pullen

Trader

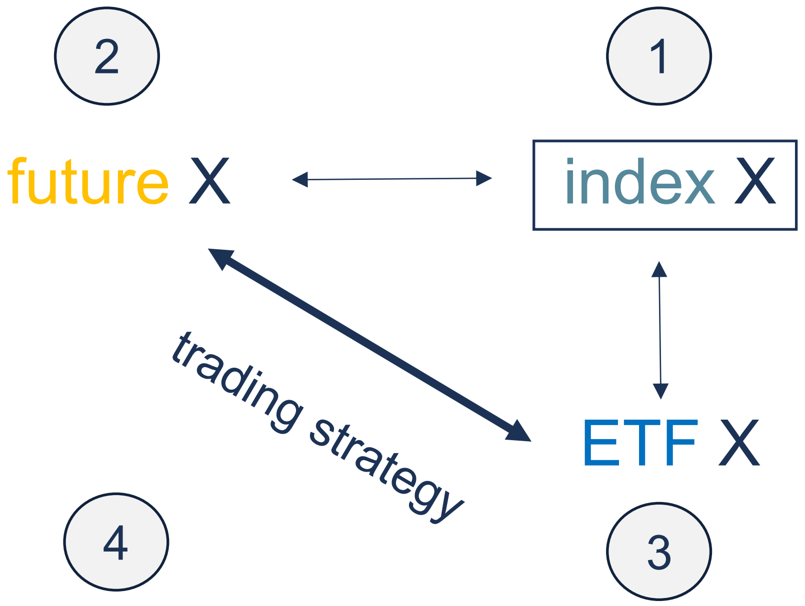
Trading trainer

Head of academic partnerships

Financial markets and instruments



Financial markets and instruments



Agenda

Index

- What is an index?
- Calculation methods
- Capitalization-weighted index value
- More about the index value

What is an index?

Basket of assets, e.g. stocks

Usually based on:

- Country or region (e.g. US, Europe, the Netherlands)
- Sector (e.g. banking, SX7E)
- Currency (e.g. Euro, SX5E)
- Capitalization (e.g. midcap, AMX)
- Theme (e.g. Linux, ecological, social)

Calculation methods

Two main methods:

- Price-weighted
  (e.g. Dow Jones and Nikkei 225)
  → only based on stock prices

- Capitalization-weighted
  (e.g. S&P500, SX5E and AEX)
  → besides stock prices, also size of companies is taken into account

Capitalization-weighted index value

Example

$$X_t = \sum_{i=1}^{n} \frac{w_i * S_{i,t}}{divisor}$$

where:

X_t is index value at time t
$S_{i,t}$ is stock i's value at time t
w_i is the number of stocks S_i in the index basket

Stock name (i)	Stock price ($S_{i,t}$)	# of shares (w_i)	Market cap ($w_i * S_{i,t}$)
Stock A	15.35	4.54 bn	69.689 bn
Stock B	14.28	3.90 bn	55.692 bn
Stock C	161.55	0.56 bn	90.468 bn

Total market cap of the index: **215.849 bn**

Divisor 63,485,000 ÷

Index value (X_t) **3400**

More about the index value

- Free float adjustment
  free float is mostly determined by excluding strategic holdings (which are more for control purposes)

- Cap on weighting of individual stocks
  a maximum percentage weighting of a stock

- Last price / bid / ask of underlying stocks
  index value based on last prices of stocks, which are sometimes not recently updated and can be old information

Futures

- What is a future?
- Underlying assets
- Value of a futures contract
- Cost of Carry
- Financing costs
- Continuously compounded interest
- Future value
- Exchange for Physical (EFP)

What is a future?

Agreement to buy (or sell) an asset at a certain moment in the future at a certain price

Traded on exchange

Standardized features of contract specified by exchange

Underlying assets

Oil & energy

Metal

Grain & oilseed

Food & fiber

Cattle & hogs

Currency

Interest

Index

Stock

Volatility

Value of a futures contract

$$F_t = S_t + \text{cost of carry}$$

<u>where:</u>

F_t = Future value at time t

S_t = Spot value at time t = current price = present value (PV)

Cost of carry = ?

Cost of Carry

‘costs’ of carry:

- **Financing cost**
- Warehousing / storage cost
- Insurance cost
- Spoilage
- Transportation cost
- Dividends

now

time

moment in the future
(expiration date)

Financing cost

?

1 year 4%

simple interest

104

now

future date

present value (S_t)
of money

future value (F_t)
of money

Continuously compounded interest

Question:

What is the future value of 1000 GBP after 3 months with a continuously compounded interest rate of 4%?

Use:

$$\mathbf{F_t = S_t * e^{(r*\tau)}}$$

where:

F_t = future value at time t
S_t = spot value at time t
r = interest rate (annual)
$\boldsymbol{\tau}$ = time to maturity (in years)

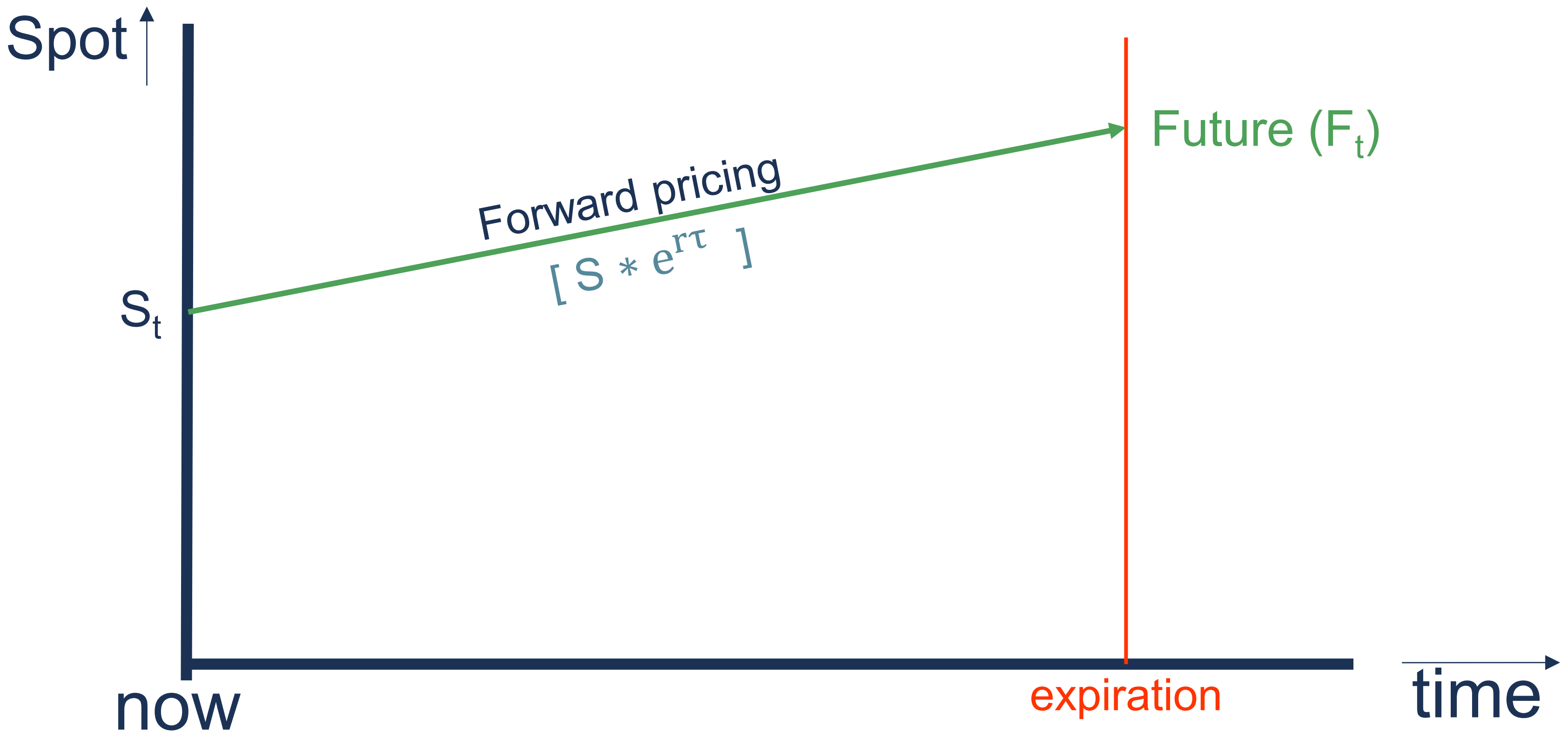
Answer:

S_t = 1000
r = 0.04
$\boldsymbol{\tau}$ = 3/12 = 0.25

F_t = $\mathbf{1000 * e^{(0.04*0.25)}}$ = 1010.05

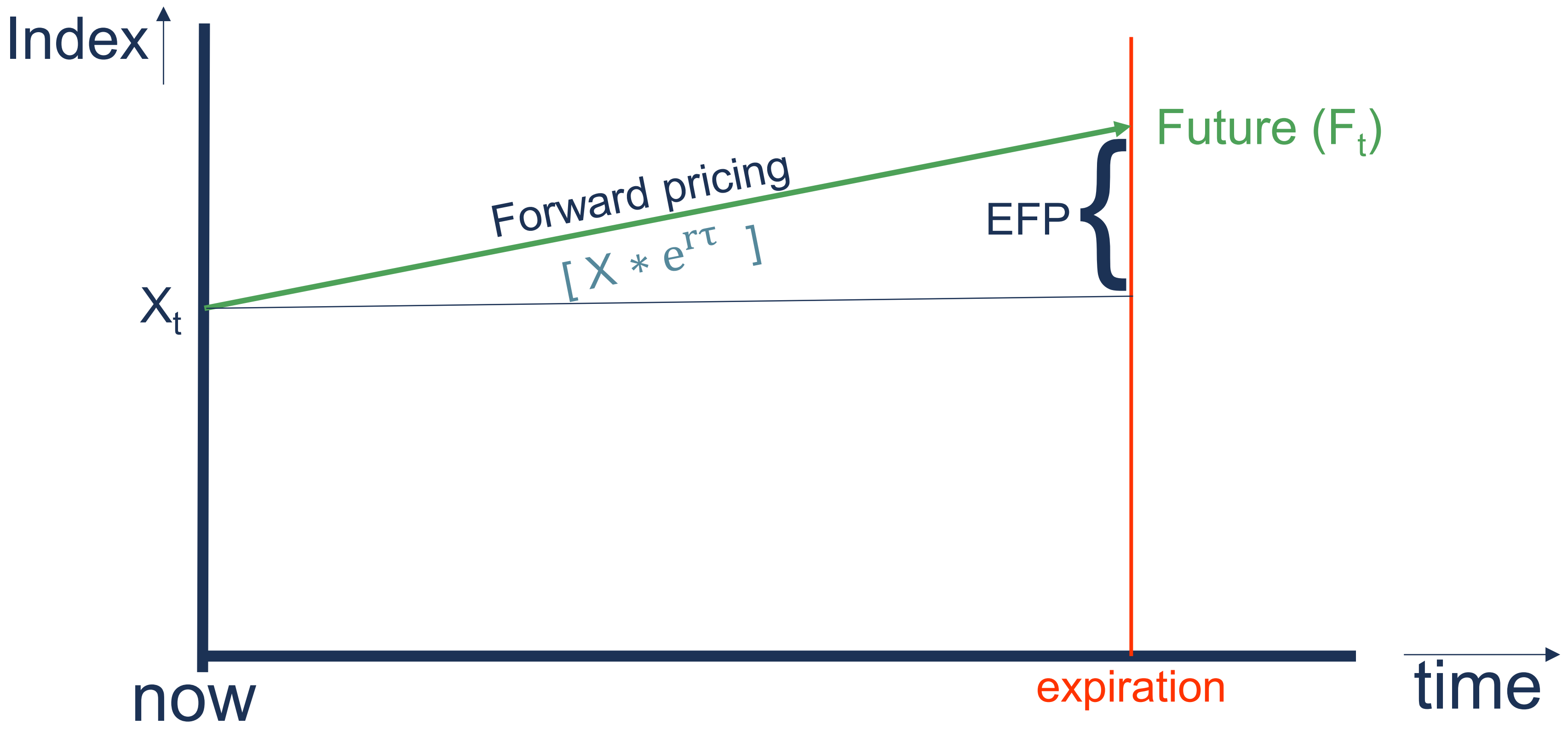
Future value

Note: we assume here that stocks do not pay dividends!



Exchange for Physical (EFP)

Note[1]: we assume here that stocks do not pay dividends!



EFP = ($F_t - X_t$)

Note[2]: an EFP only refers to a stocks index future

ETFs

- What is an ETF?
- Multiplier
- Cash Component
- Net Asset Value (NAV)
- ETF bid/ask price
- Market Making in ETFs
- Hedge

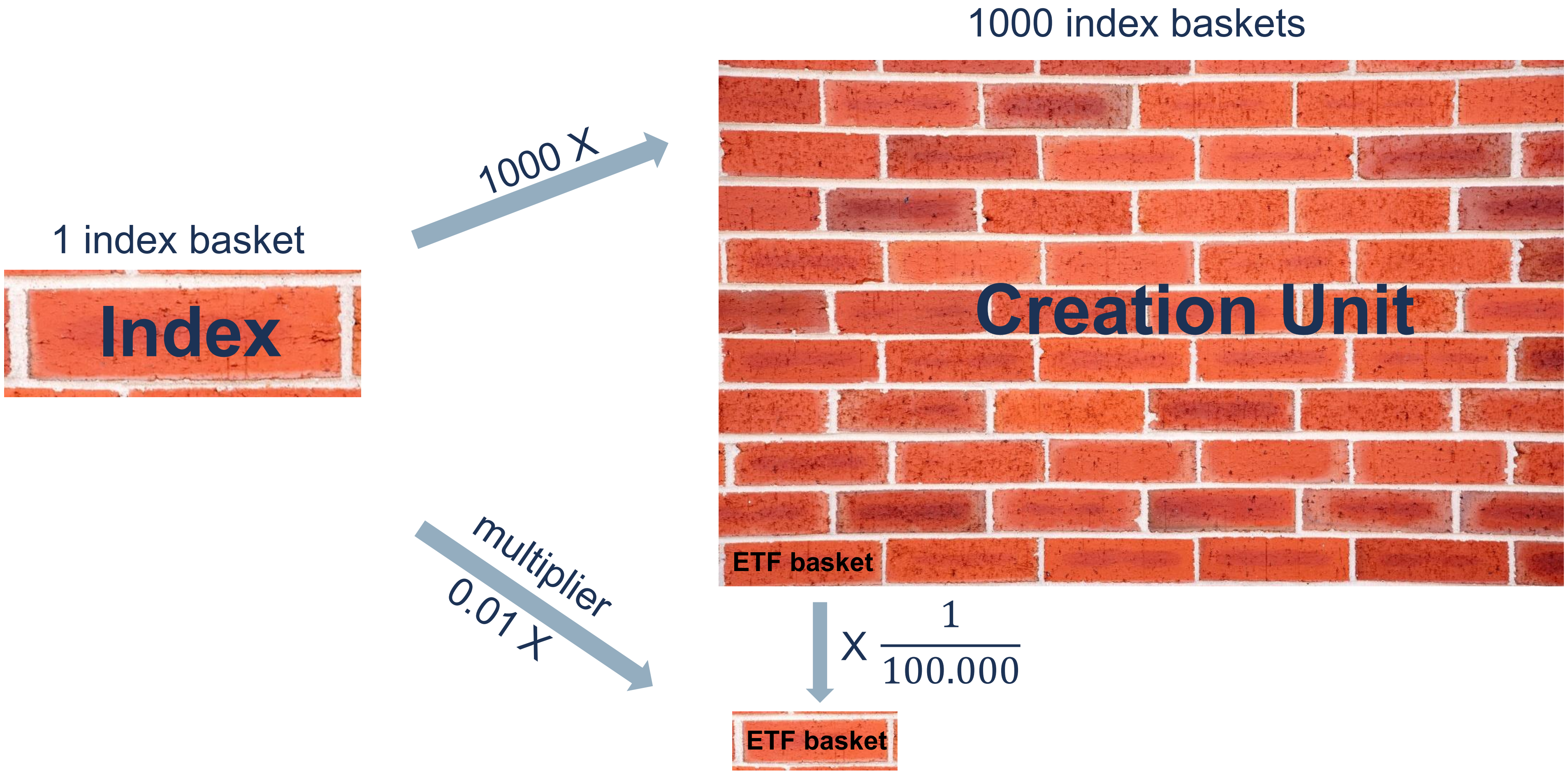
What are ETFs?

Exchange-traded funds (ETFs) are investment entities that offer investors a way to pool their money in a fund that invests in stocks, bonds, or other assets.
In return, investors receive an interest in the fund.

Characteristics:

- **E**xchange **T**raded **F**und
- (index) trackers
- access to anything
- intraday trading
- low costs
- transparent
- tax efficient

Multiplier



1000 index baskets
1 index basket
Index
1000 X
Creation Unit
ETF basket
multiplier
0.01 X
X $\frac{1}{100.000}$
ETF basket

Cash Component

Net Asset Value (NAV)

Example

$\mathrm{NAV} = \mathrm{C} + \mathrm{M} * \mathrm{X_t}$

where:

NAV_t is the Net Asset Value of the ETF at time t
X_t is index value at time t
M is the multiplier of the ETF
C is the cash component of the ETF

NAV
43.52

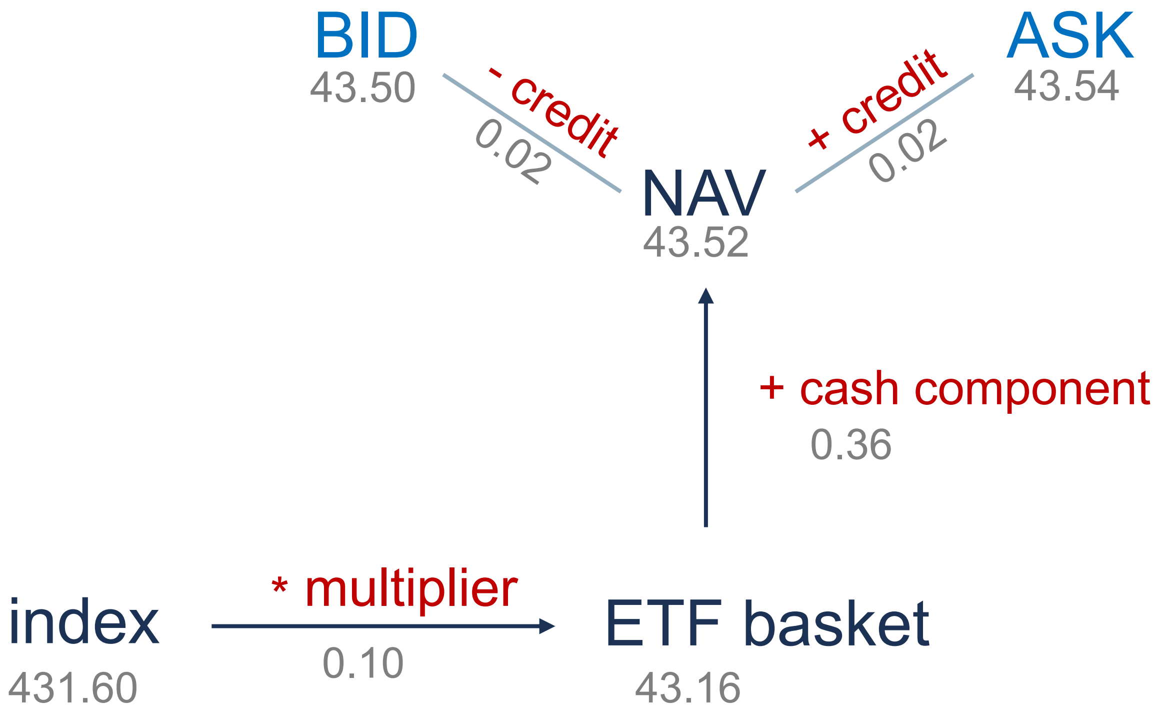
+ cash component
0.36

index
431.60

* multiplier
0.10

ETF basket
43.16

ETF bid/ask price



Credit is the targeted amount of profit when doing the trade

Market Making in ETFs

bid #	price	ask #
	43.56	95200
	43.55	85000
	43.54	**71200**
	43.53	55320
	43.52	20090
	43.51	15000
15015	**43.50**	
25040	43.49	
50000	43.48	
60100	43.47	
80030	43.46	
96150	43.45	

BID 43.50 — your MM quote on the screen — ASK 43.54

After an ETF trade, how do you hedge?

A) Wait until you can do an opposite trade in the ETF

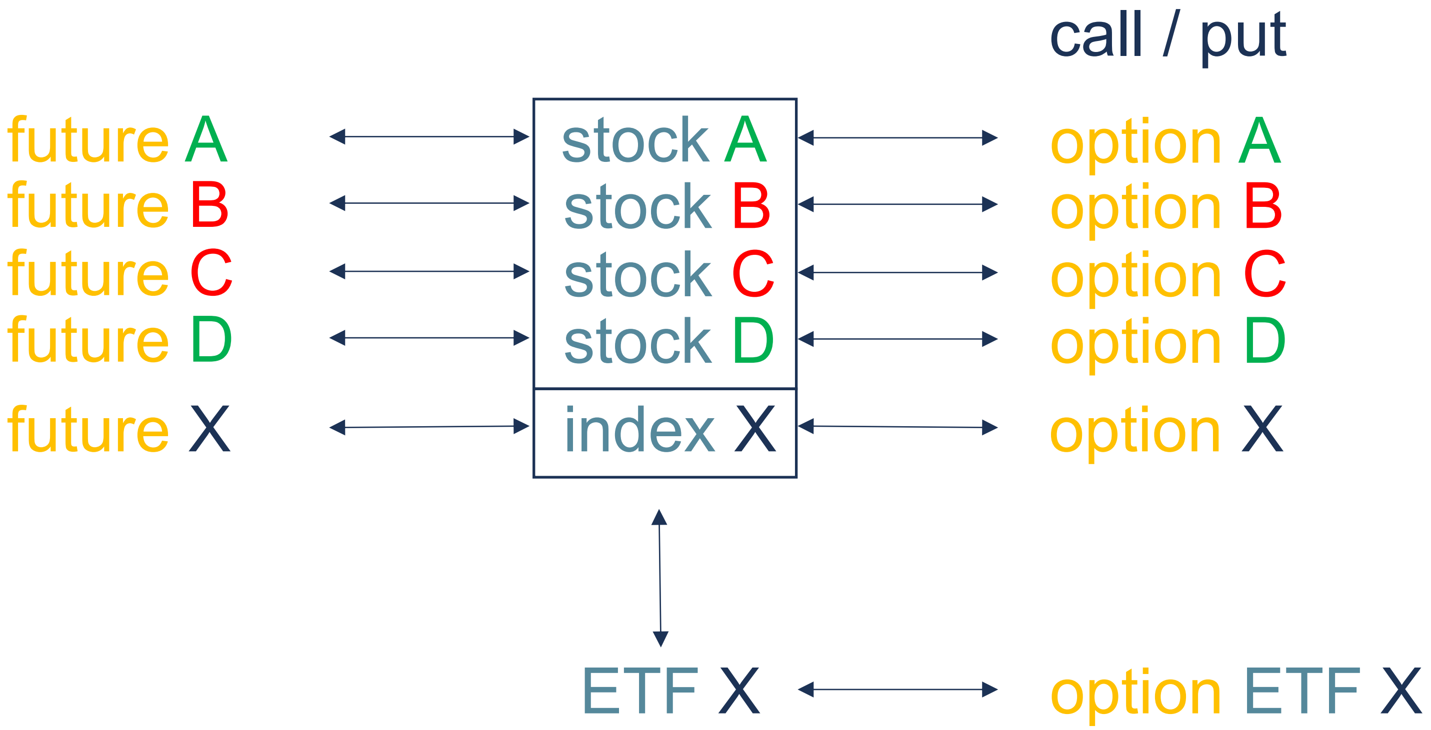
B) Trade all stocks of the underlying basket, i.e. trade the index

C) Trade a future on the index (if there is a suitable future available)

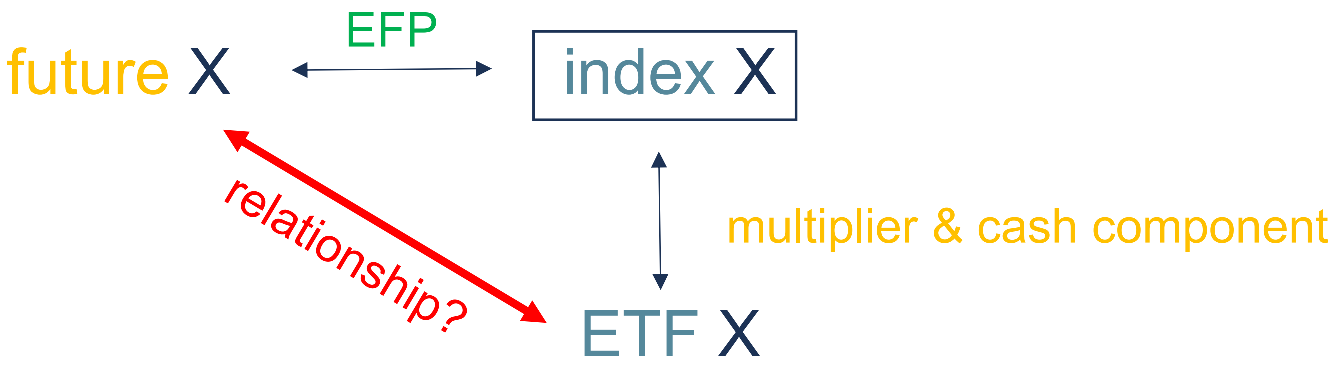
Trading strategy

- Future versus ETF
- Relationship futures and ETFs
- Market Making and Hedging
- Example
- Exercise & Answer

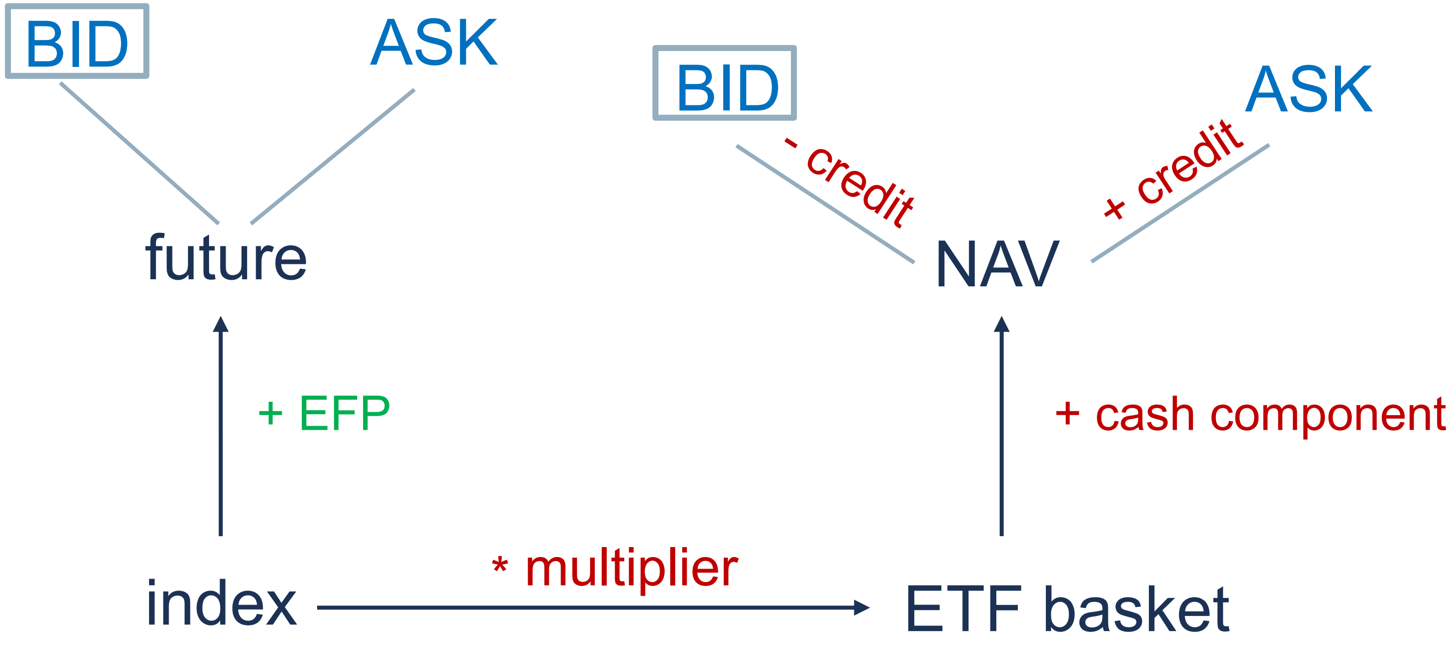
Financial markets and instruments



Futures vs ETFs



Relationship futures and ETFs



Example

Calculation of the ETF bid price

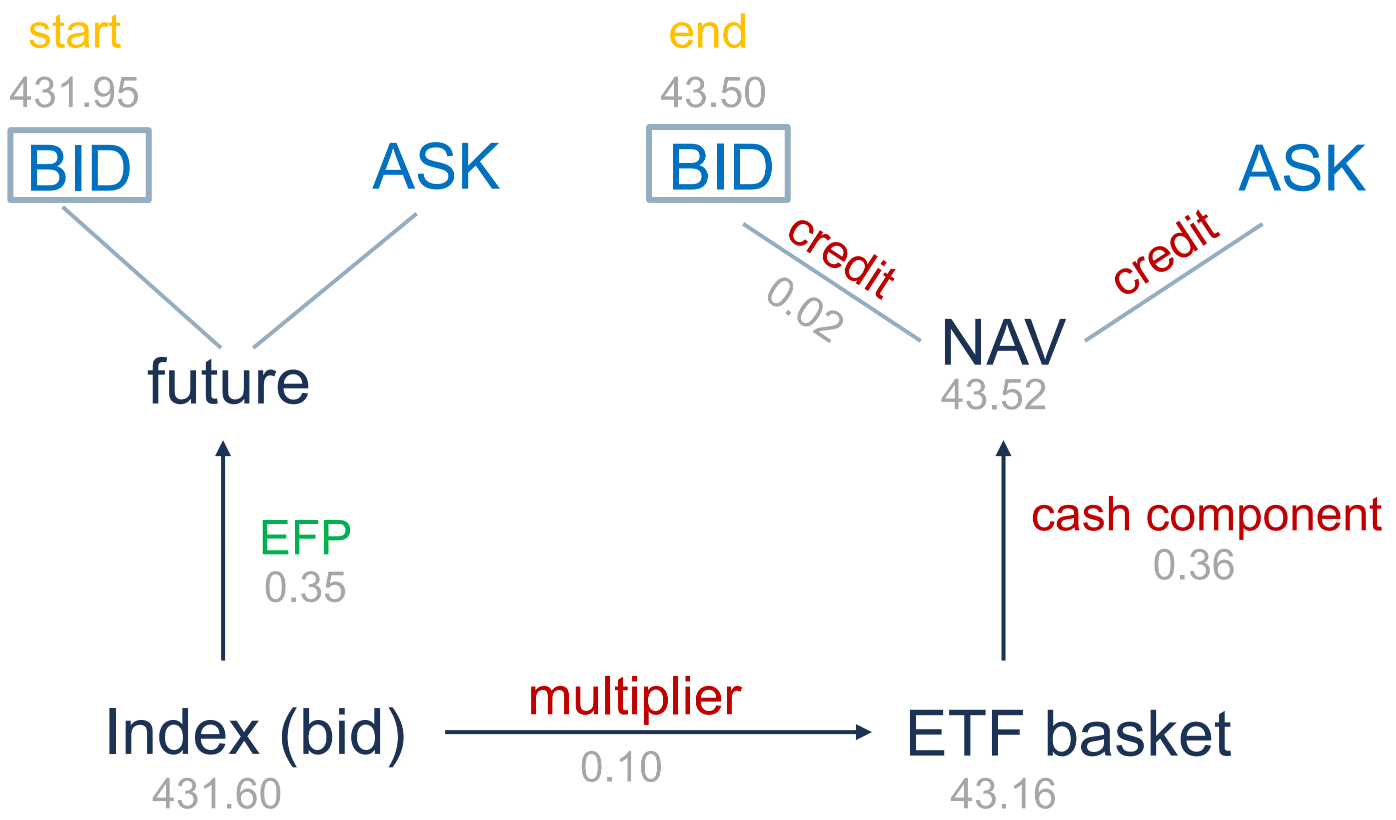
Future best bid

bid #	price	ask #
	432.25	902
	432.20	824
	432.15	697
	432.10	520
	432.05	430
	432.00	210
110	431.95	
323	431.90	
457	431.85	
680	431.80	
890	431.75	
970	431.70	

-EFP * multiplier + CC - credit = ETF bid

$$\underbrace{\underbrace{\underbrace{\underbrace{[\,431.95 - 0.35\,]}_{\text{index (bid)}} * 0.10}_{\text{ETF basket (bid)}} + 0.36}_{\text{NAV (bid)}} - 0.02}_{\text{ETF bid}} = \underline{43.50}$$

Example calculation bid price ETF



Futures vs ETFs

hedge

buy ETFs → sell futures

1) if buy @

bid #	price	ask #
	43.56	95200
	43.55	85000
	43.54	**71200**
	43.53	55320
	43.52	20090
	43.51	15000
15015	$\boxed{\textbf{43.50}}$	
25040	43.49	
50000	43.48	
60100	43.47	
80030	43.46	
96150	43.45	

2) then sell @

bid #	price	ask #
	432.25	902
	432.20	824
	432.15	697
	432.10	520
	432.05	430
	432.00	210
110	$\boxed{431.95}$	
323	431.90	
457	431.85	
680	431.80	
890	431.75	
970	431.70	

3) a profit of 0.02 (i.e. the credit) has been made

Exercise

Make a market (both bid and ask) for an ETF on the Eurostoxx 50 index, based on the following parameters:

future best bid = 2852

future best ask = 2853

EFP = 5.00

multiplier = 0.01

cash component = 0.21

credit = 0.02

Round your bid & ask prices to 2 decimals

Answer

	Bid	Ask
future	2852	2853
EFP	5.00	5.00
index	2847	2848
multiplier	0.01	0.01
EFT basket	28.47	28.48
cash comp.	0.21	0.21
NAV	28.68	28.69
credit	0.02	0.02
market	28.66	28.71

If necessary, bids have to be rounded down to 2 decimals,
asks have to be rounded up to 2 decimals

Thank you!

Robbert Pullen
Trading trainer

RobbertPullen@optiver.com
Strawinskylaan 3095
1077 ZX Amsterdam